MACHINE LEARNING

PRACTICAL COURSE

SUPERVISED LEARNING

OVER FITTING , UNDER FITTING
REGULARIZATION
LOGISTIC REGRESSION
EVALUATION

SUPERVISED LEARNING

OVER FITTING , UNDER FITTING

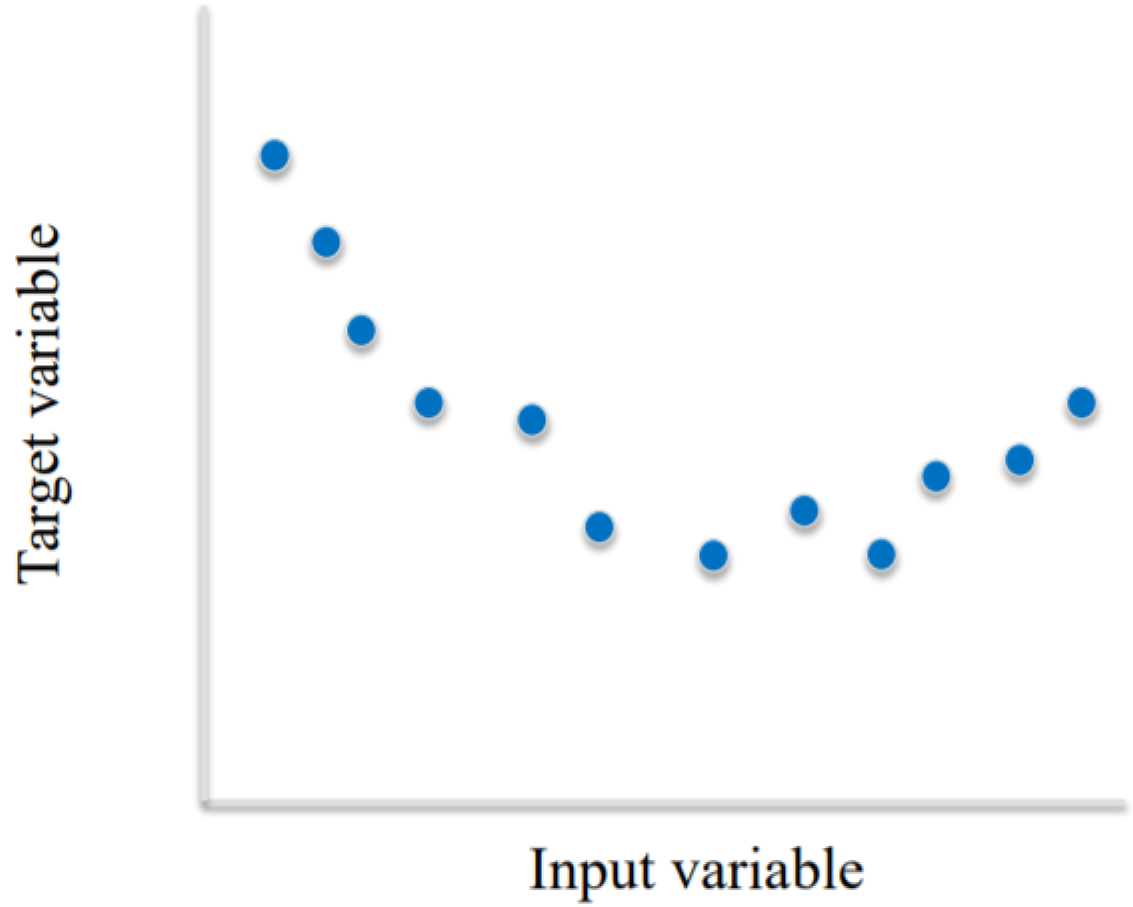
MACHINE LEARNING BY PROBLEM TYPE

- Regression
- Classification
- Dimension reduction
- Clustering
- Control

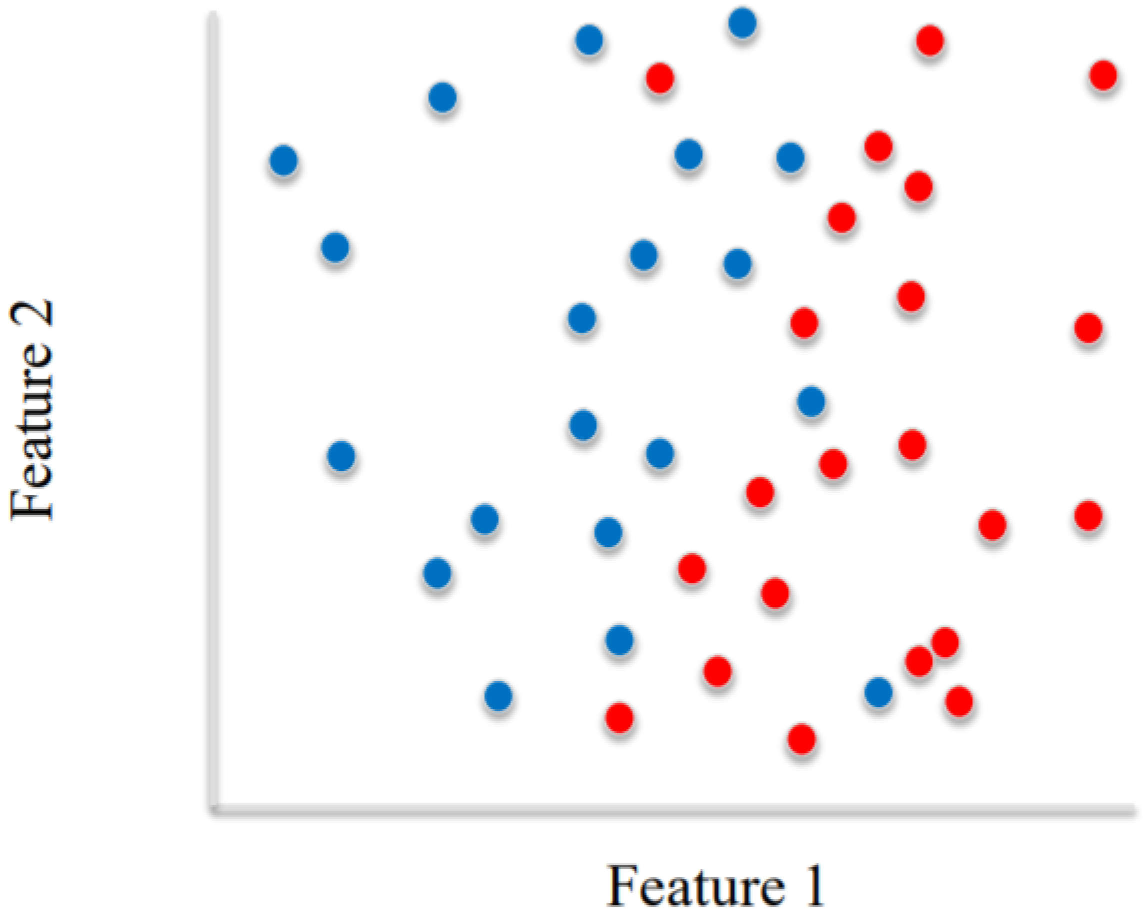
OVER FITTING, UNDER FITTING

- Generalization ability refers to an algorithm's ability to give accurate predictions for new, previously unseen data.
- Assumptions:
  - *Future unseen data (test set) will have the same properties as the current training sets.*
  - *Thus, models that are accurate on the training set are expected to be accurate on the test set.*
  - *But that may not happen if the trained model is tuned too specifically to the training set.*
- Models that are too complex for the amount of training data available are said to overfit and are not likely to generalize well to new examples.
- Models that are too simple, that don't even do well on the training data, are said to underfit and also not likely to generalize well.

Overfitting in regression



Overfitting in classification



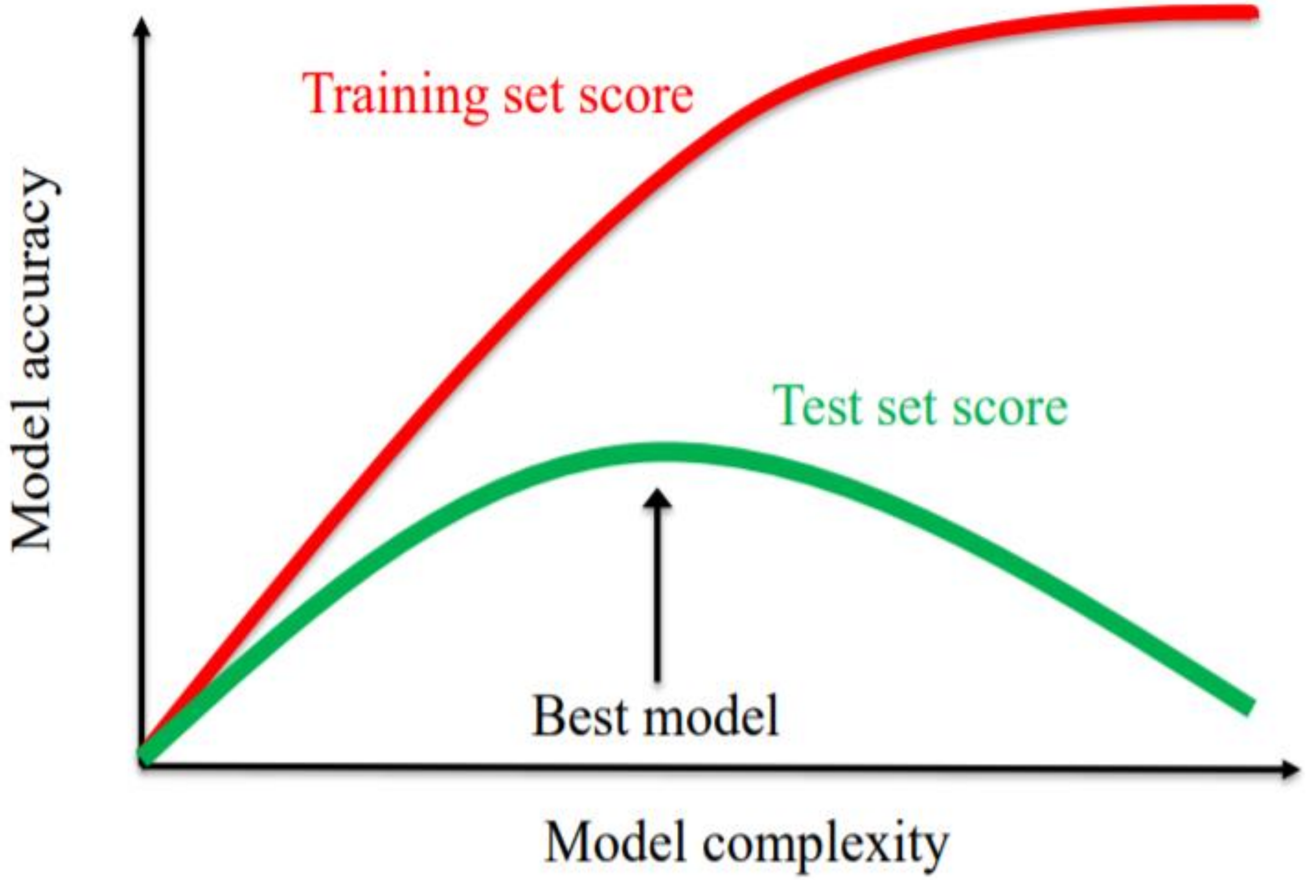
SUPERVISED LEARNING

METHODS

SUPERVISED LEARNING METHODS

- **We'll cover a number of widely-used supervised learning methods for classification and regression.**
- **For each supervised learning method we'll explore:**
  - *How the method works conceptually at a high level.*
  - *What kind of feature preprocessing is typically needed.*
  - *Key parameters that control model complexity, to avoid under- and over-fitting.*
  - *Positives and negatives of the learning method.*

The relationship between model complexity and training/test performance



LINEAR MODELS

- A linear model is a sum of weighted variables that predicts a target output value given an input data instance. Example: *predicting housing prices*
  - *House features: taxes per year (X_{TAX}), age in years (X_{AGE})*

$$\widehat{Y_{PRICE}} = 212000 + 109\, X_{TAX} - 2000\, X_{AGE}$$

  - *A house with feature values (X_{TAX}, X_{AGE}) of (10000, 75) would have a predicted selling price of:*

$$\widehat{Y_{PRICE}} = 212000 + 109 \cdot \mathbf{10000} - 2000 \cdot \mathbf{75} = \mathbf{1{,}152{,}000}$$

Linear Regression is an Example of a Linear Model

Input instance – feature vector: $\boldsymbol{x} = (x_0, x_1, \ldots\ , x_n)$

Predicted output: $\hat{y} = \widehat{w_0} x_0 + \widehat{w_1} x_1 + \cdots \widehat{w_n} x_n + \hat{b}$

Parameters to estimate:

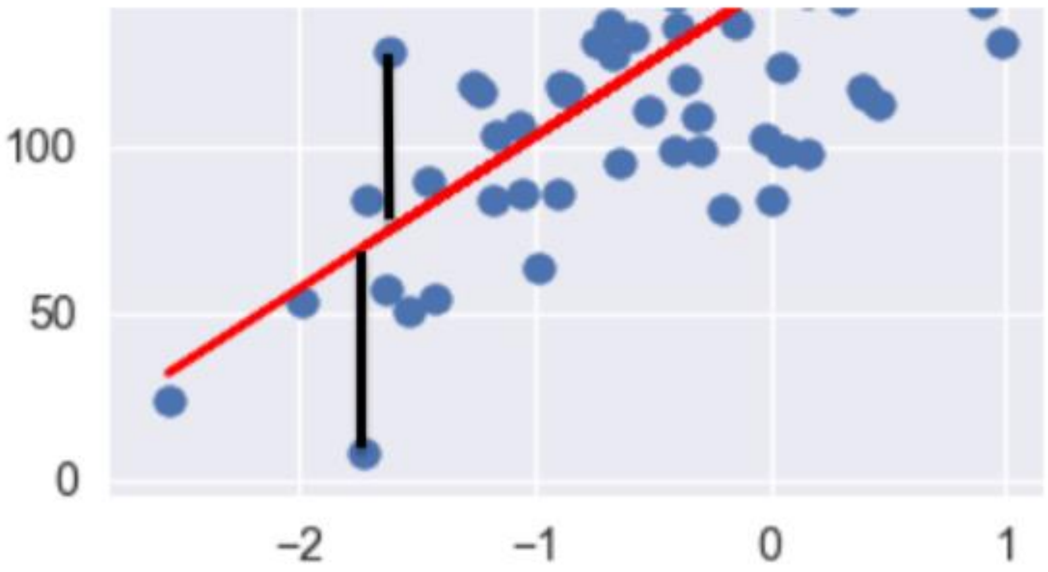
- $\hat{\boldsymbol{w}} = (\widehat{w_0}, \cdots, \widehat{w_n})$: feature weights/ model coefficients
- $\hat{\boldsymbol{b}}$: constant bias term / intercept

How are Linear Regression Parameters *w*, *b* Estimated?

- **Parameters are estimated from training data.**
- **There are many different ways to estimate *w* and *b*:**
  - *Different methods correspond to different "fit" criteria and goals and ways to control model complexity.*
- **The learning algorithm finds the parameters that optimize an objective function, typically to minimize some kind of loss function of the predicted target values *vs.* actual target values.**

Least-Squares Linear Regression ("Ordinary least-squares")

- Finds *w* and *b* that minimizes the sum of squared differences (RSS) over the training data between predicted target and actual target values.
- a.k.a. mean squared error of the linear model
- No parameters to control model complexity.



$$RSS(\boldsymbol{w}, b) = \sum_{\{i=1\}}^{N} (\boldsymbol{y}_i - (\boldsymbol{w} \cdot \boldsymbol{x}_i + b))^2$$

OVER FITTING SOLUTIONS

- Based on data
- Based on features
- Based on model type (neural networks/ KNN/ SVM..)
- Based cost function
- Based on training process
- Based on model architecture

OVER FITTING SOLUTIONS

- Based on data (add more data)
- Based on features (feature selection)
- Based on model type (neural networks dropout/ KNN/ SVM..)
- Based cost function (regularization)
- Based on training process (early stopping, checkpoints,..)
- Based on model architecture (less layers, less forests, change model)

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REGULARIZATION

Ridge Regression

- Ridge regression learns *w*, *b* using the same least-squares criterion but adds a penalty for large variations in *w* parameters

$$RSS_{RIDGE}(\boldsymbol{w}, b) = \sum_{\{i=1\}}^{N} (\boldsymbol{y}_i - (\boldsymbol{w} \cdot \boldsymbol{x}_i + b))^2 \boxed{+ \alpha \sum_{\{j=1\}}^{p} w_j^2}$$

- Once the parameters are learned, the ridge regression prediction formula is the same as ordinary least-squares.
- The addition of a parameter penalty is called regularization. Regularization prevents overfitting by restricting the model, typically to reduce its complexity.
- Ridge regression uses L2 regularization: minimize sum of squares of *w* entries
- The influence of the regularization term is controlled by the α parameter.
- Higher alpha means more regularization and simpler models.

REGULARIZATION

Lasso regression is another form of regularized linear regression that uses an L1 regularization penalty for training (instead of ridge's L2 penalty)

- **L1 penalty: Minimize the sum of the absolute values of the coefficients**

$$RSS_{LASSO}(\boldsymbol{w}, b) = \sum_{\{i=1\}}^{N} (\boldsymbol{y}_i - (\boldsymbol{w} \cdot \boldsymbol{x}_i + b))^2 \boxed{+ \alpha \sum_{\{j=1\}}^{p} |w_j|}$$

- **This has the effect of setting parameter weights in *w* to zero for the least influential variables. This is called a sparse solution: a kind of feature selection**
- **The parameter α controls amount of L1 regularization (default = 1.0).**
- **The prediction formula is the same as ordinary least-squares.**
- **When to use ridge vs lasso regression:**
  - *Many small/medium sized effects: use ridge.*
  - *Only a few variables with medium/large effect: use lasso.*

REGULARIZATION

The differences between L1 and L2 regularization:

- L1 regularization penalizes the sum of absolute values of the weights, whereas L2 regularization penalizes the sum of squares of the weights.
- The L1 regularization solution is sparse. The L2 regularization solution is non-sparse.
- L2 regularization doesn’t perform feature selection, since weights are only reduced to values near 0 instead of 0. L1 regularization has built-in feature selection.
- L1 regularization is robust to outliers, L2 regularization is not.

- **When to use ridge vs lasso regression:**
  - Many small/medium sized effects: use ridge.
  - Only a few variables with medium/large effect: use lasso.

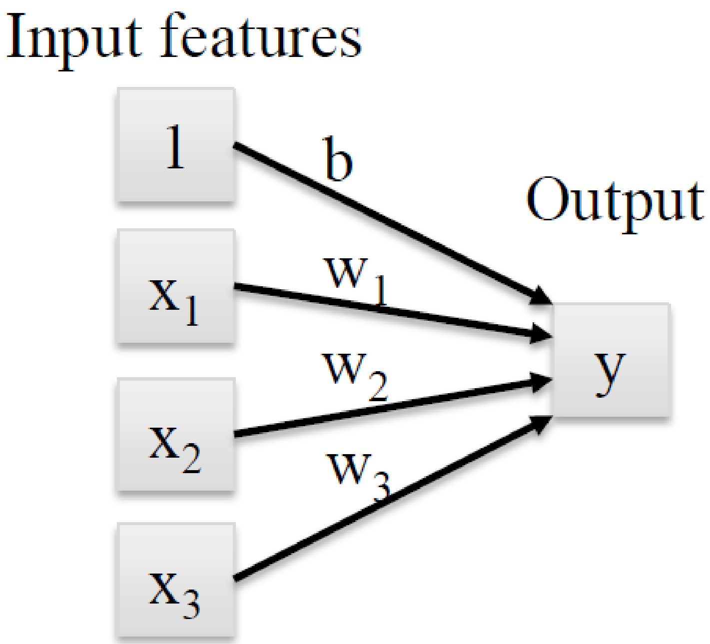
FEATURE NORMALIZATION

The Need for Feature Normalization

- **Important for some machine learning methods that all features are on the same scale (e.g. faster convergence in learning, more uniform or 'fair' influence for all weights)**
  - *e.g. regularized regression, k-NN, support vector machines, neural networks, …*
- **Can also depend on the data. More on feature engineering later in the course. For now, we do MinMax scaling of the features:**
  - *For each feature x_i : compute the min value x_i^{MIN} and the max value x_i^{MAX} achieved across all instances in the training set.*
  - *For each feature: transform a given feature x_i value to a scaled version x_i' using the formula*

$$x_i' = (x_i - x_i^{MIN})/(x_i^{MAX} - x_i^{MIN})$$

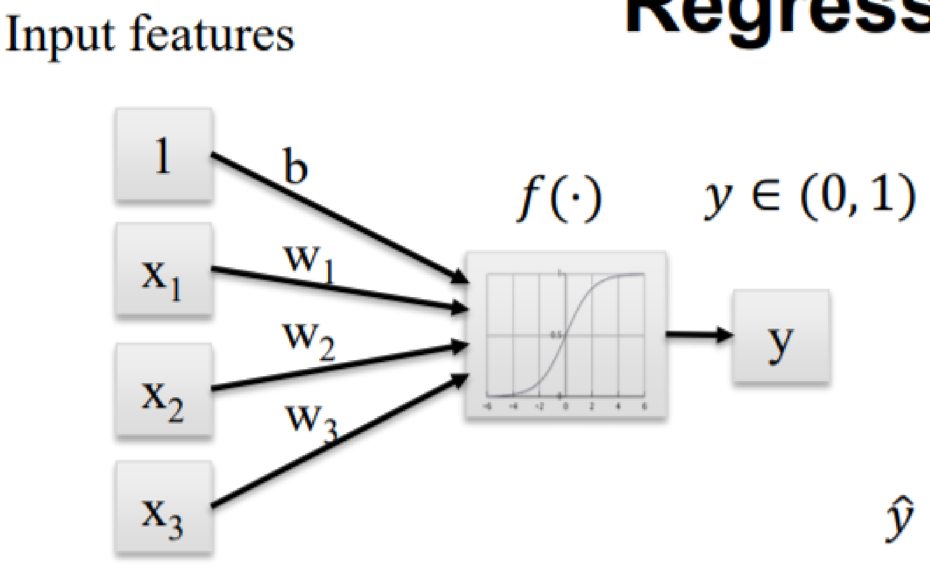
Linear Regression

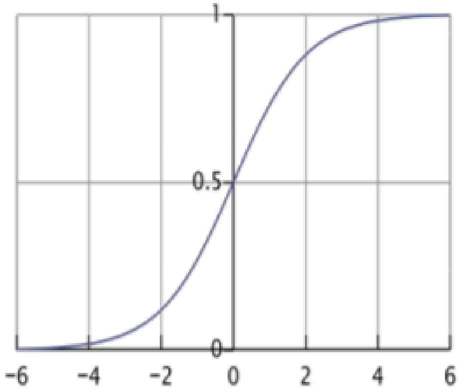


$$\hat{y} = \hat{b} + \widehat{w}_1 \cdot x_1 + \cdots \widehat{w}_n \cdot x_n$$

LOGISTIC REGRESSION

Linear models for classification: Logistic Regression



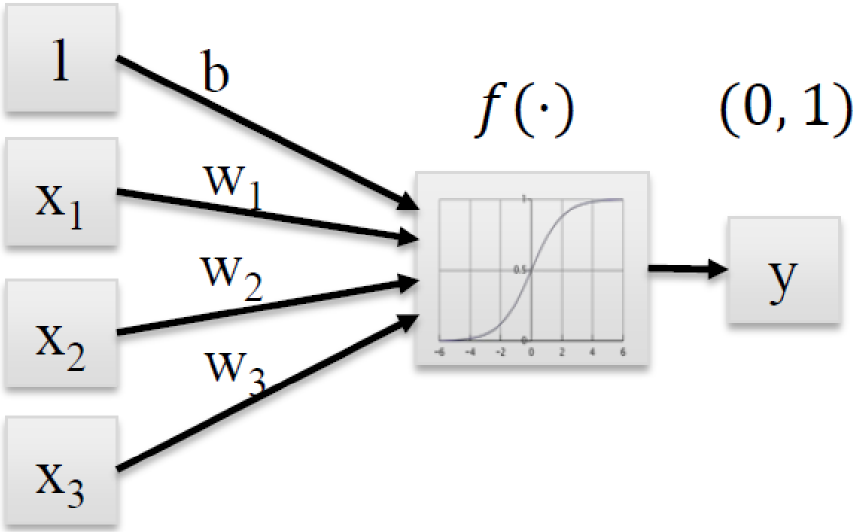


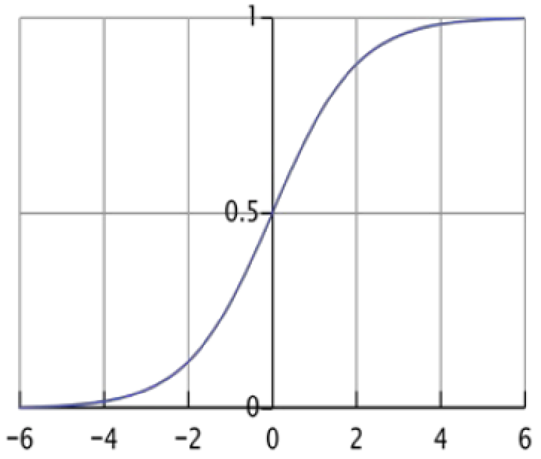
$$\hat{y} = \text{logistic}(\hat{b} + \hat{w}_1 \cdot x_1 + \cdots \hat{w}_n \cdot x_n)$$

$$= \frac{1}{1 + \exp\left[-(\hat{b} + \hat{w}_1 \cdot x_1 + \cdots \hat{w}_n \cdot x_n)\right]}$$

Linear models for classification: Logistic Regression

Input features



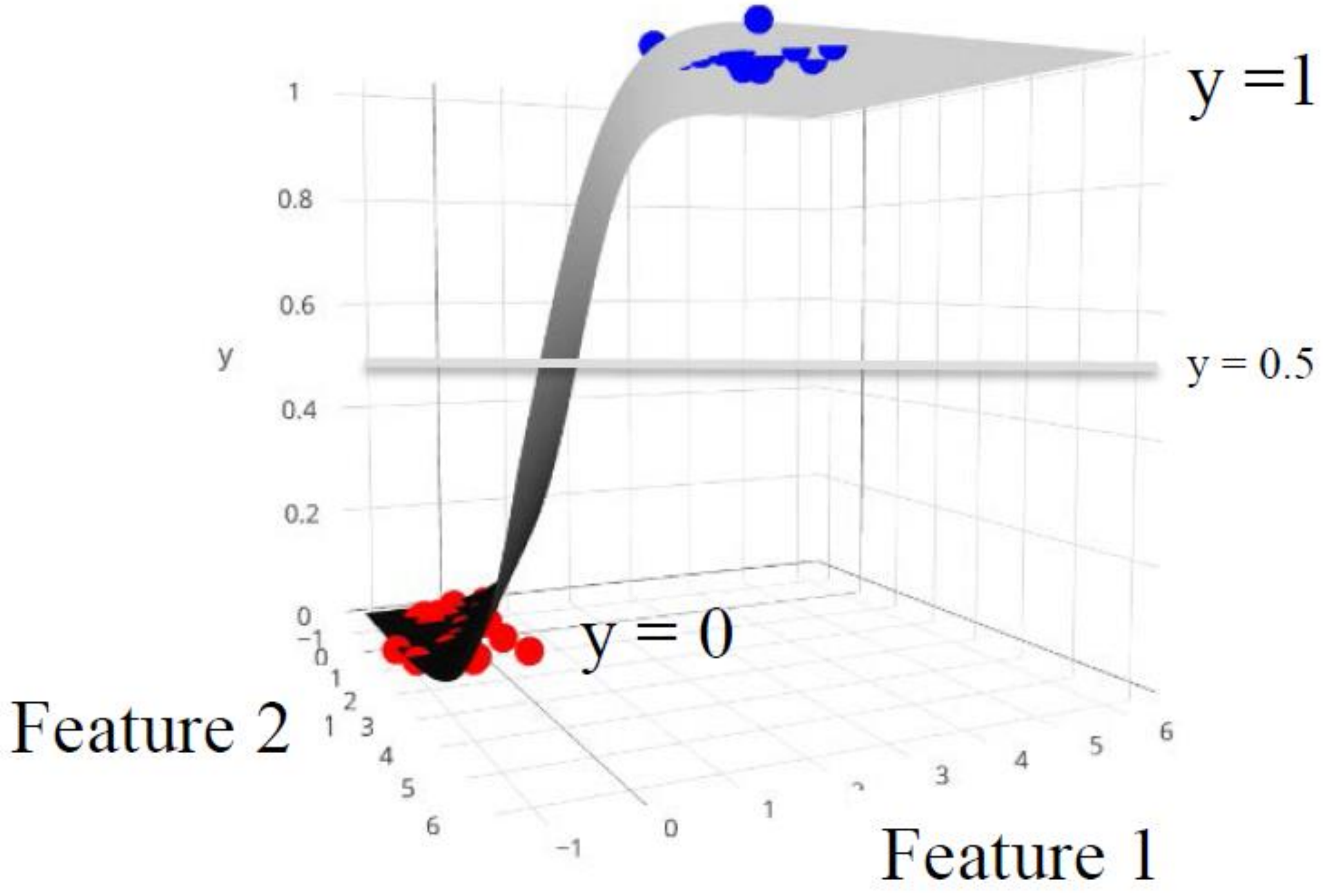


The logistic function transforms real-valued input to an output number y between 0 and 1, interpreted as the probability the input object belongs to the positive class, given its input features $(x_0, x_1, \dots, x_n)$

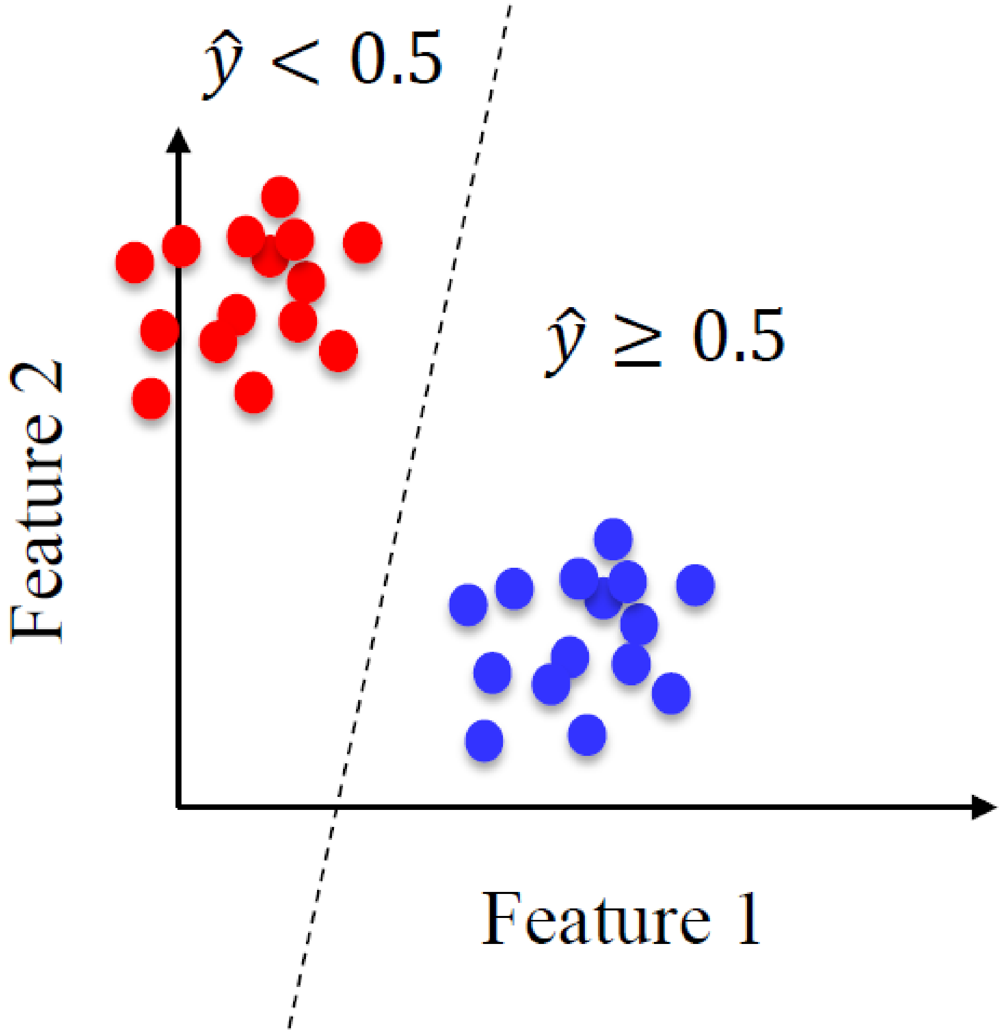
$$\hat{y} = \text{logistic}(\hat{b} + \hat{w}_1 \cdot x_1 + \cdots \hat{w}_n \cdot x_n)$$

$$= \frac{1}{1 + \exp\left[-(\hat{b} + \hat{w}_1 \cdot x_1 + \cdots \hat{w}_n \cdot x_n)\right]}$$

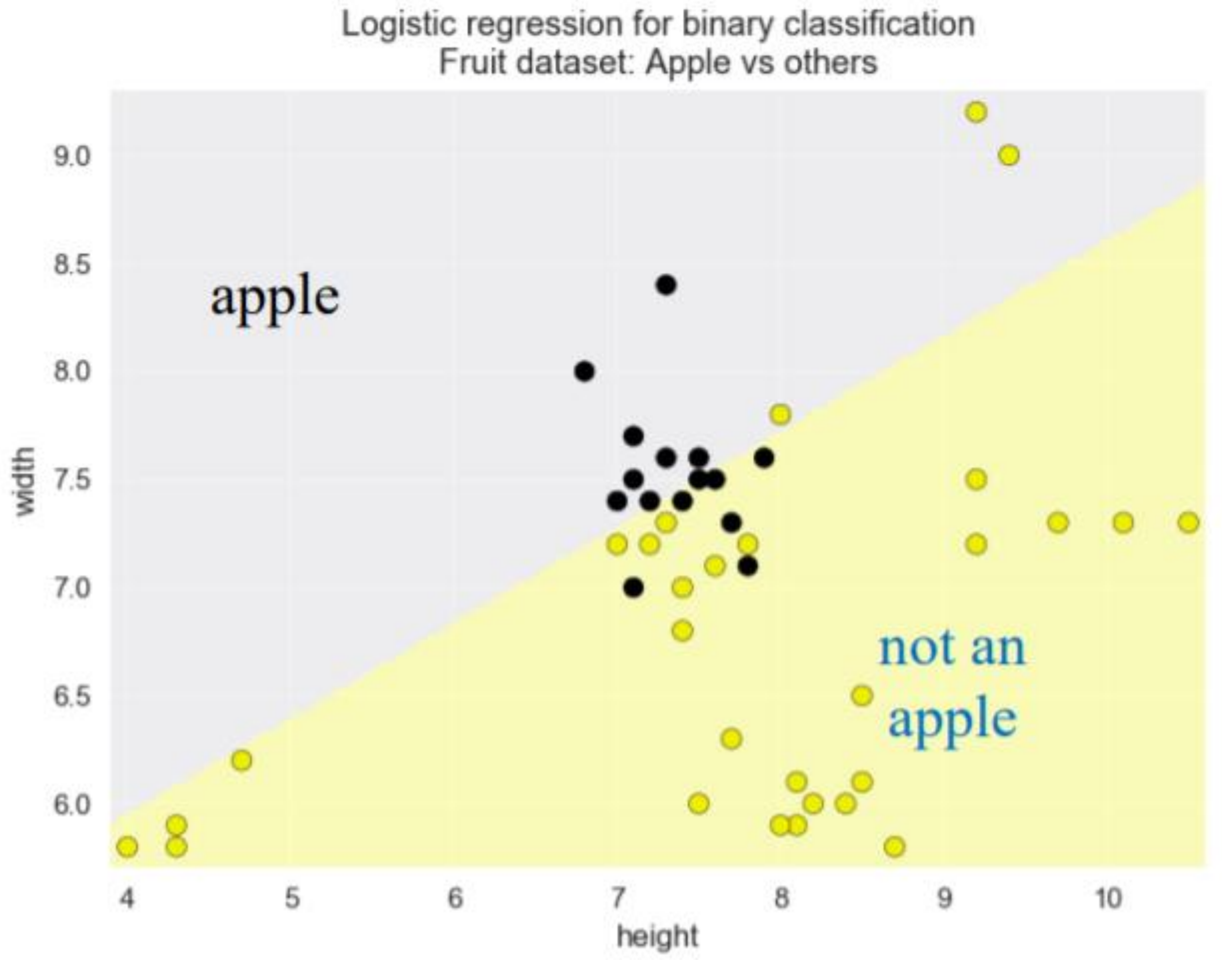
Logistic Regression for binary classification



Logistic Regression for binary classification

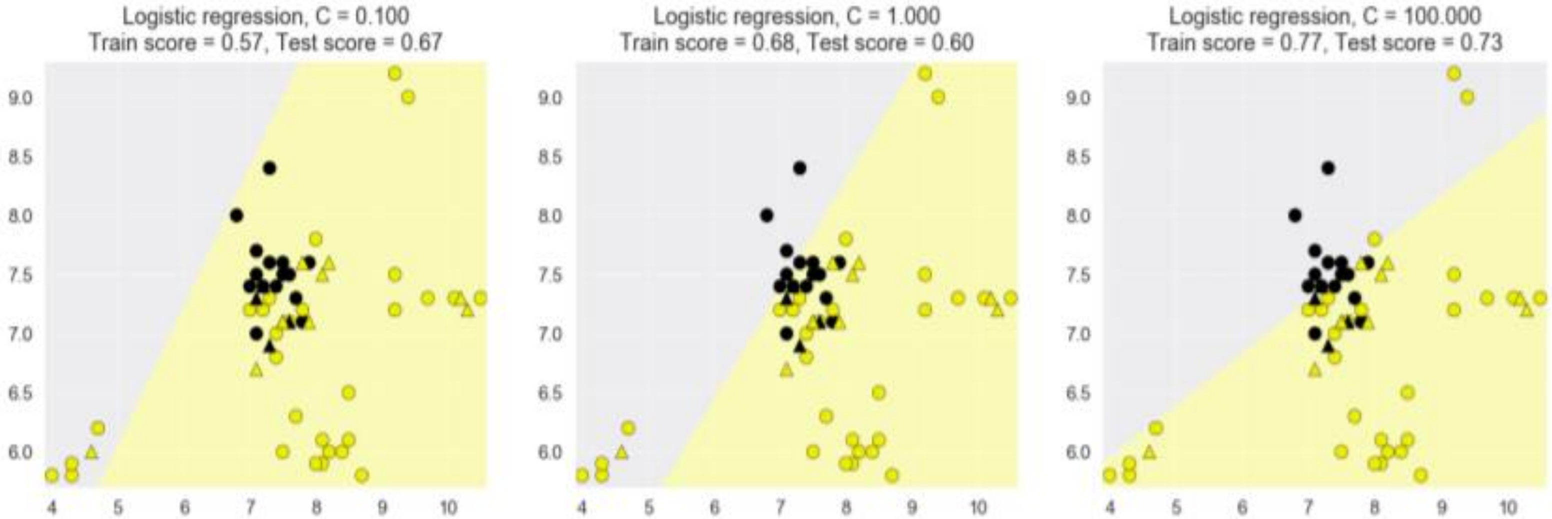


Simple logistic regression problem: two-class, two-feature version of the fruit dataset



Logistic Regression: Regularization

- L2 regularization is 'on' by default (like ridge regression)
- Parameter C controls amount of regularization (default 1.0)
- As with regularized linear regression, it can be important to normalize all features so that they are on the same scale.



C: float, default=1.0 Inverse of regularization strength; must be a positive float. Like in support vector machines, smaller values specify stronger regularization.

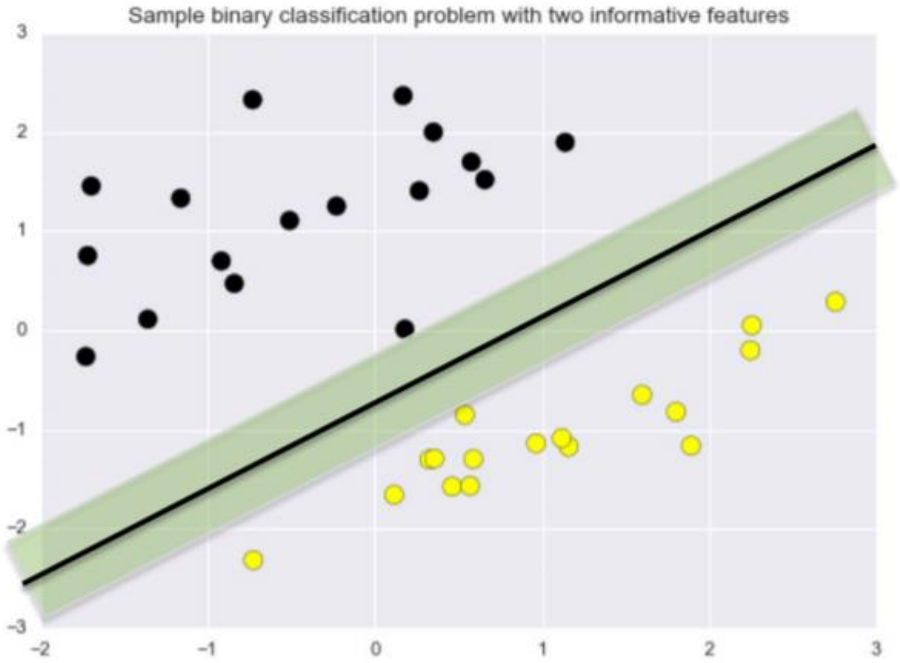
Linear Classifiers: Support Vector Machines

Maximum margin linear classifier: Linear Support Vector Machines

$$f(x, w, b) = sign(w \circ x + b)$$

<u>Maximum margin classifier</u>

The linear classifier with maximum margin is a linear Support Vector Machine (LSVM).



Regularization for SVMs: the C parameter

- **The strength of regularization is determined by C**
- **Larger values of C: less regularization**
  - *Fit the training data as well as possible*
  - *Each individual data point is important to classify correctly*
- **Smaller values of C: more regularization**
  - *More tolerant of errors on individual data points*

Linear Models: Pros and Cons

Pros:

- Simple and easy to train.
- Fast prediction.
- Scales well to very large datasets.
- Works well with sparse data.
- Reasons for prediction are relatively easy to interpret.

Cons:

- For lower-dimensional data, other models may have superior generalization performance.
- For classification, data may not be linearly separable (more on this in SVMs with non-linear kernels)

linear_model: Important Parameters

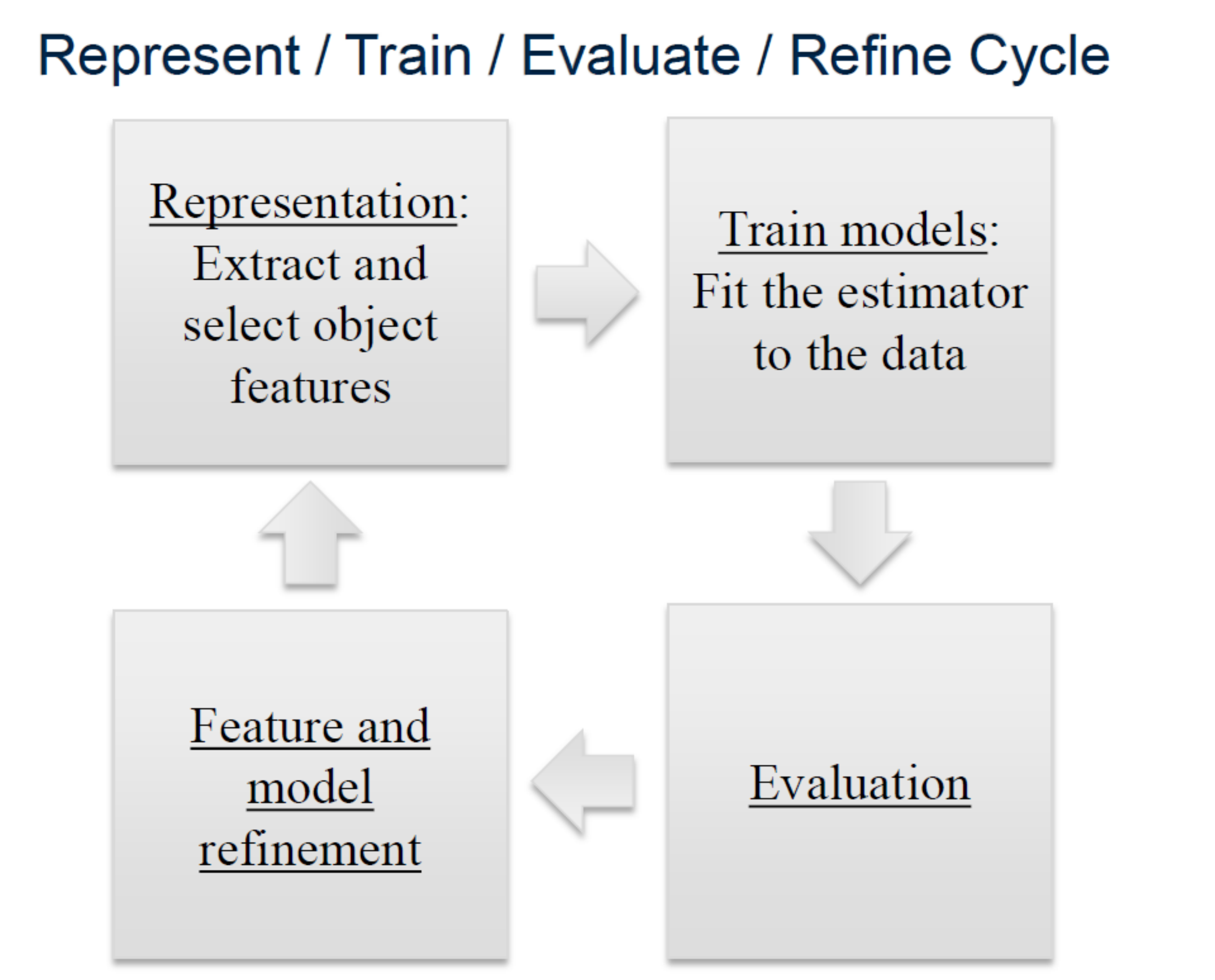
Model complexity

- **alpha: weight given to the L1 or L2 regularization term in regression models**
  - *default = 1.0*
- **C: regularization weight for LinearSVC and LogisticRegression classification models**
  - *default = 1.0*

SUPERVISED LEARNING

EVALUATION

EVALUATION


Represent / Train / Evaluate / Refine Cycle
Representation: Extract and select object features
Train models: Fit the estimator to the data
Evaluation
Feature and model refinement

EVALUATION

- Different applications have very different goals

Accuracy is widely used, but many others are possible, e.g.:

  - User satisfaction (Web search)
  - Amount of revenue (e-commerce)
  - Increase in patient survival rates (medical)

- It's very important to choose evaluation methods that match the goal of your application.
- Compute your selected evaluation metric for multiple different models.
- Then select the model with 'best' value of evaluation metric.

EVALUATION

- Evaluation of model's performance or accuracy is usually split in two ways: **offline** and **online**.
- **Offline evaluation** is done with already available data using well-known methods such as splitting of data into train and test samples, cross-validating and recording the performance of those sets. The goal is to find the performance of a model on data points that are present in databases and deploy the best possible model up to a knowledge at the time of training.
- On the other hand, **online evaluation** is done with non-existent data at a given time. The goal is to find performance of a model on data points that are not available in databases. In other words, the deployed model is affecting users we don't know but we would like to measure how well this model works for them and business overall. The common metrics of measurements are heavily influenced by the problem the model is trying to solve and/or business KPIs.
- **A good Machine Learning model should improve both on offline and online evaluation steps over time**

OFFLINE EVALUATION METRICS

For Classification:

- Accuracy
- Confusion matrix
- Precision
- Recall
- F1 Score
- AUC-ROC
- ..

For Regression :

- Mean Squared Error or MSE
- Root Mean Squared Error or RMSE
- Mean Absolute Error or MAE
- Root Mean Squared Log Error or RMSLE
- R-Squared
- Adjusted R-Squared

For Clustering:

- Dunn Index
- Elbow method
- Silhouette Coefficient

NOTEBOOKS

NOTEBOOKS

1. Supervised Learning with sikit-learn (Linear Regression + Logistic Regression + KNN)
2. Model_Evaluation